					Course Code:A2610001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ENGLISH for TECHNICAL COMMUNICATION			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Identify any four expressions for asking for permissions.	**1**	**2**	**2**
	b).	Complete the sentences with the appropriate words as directed in the brackets. i. The students are interested ___ learning new technologies. (Preposition) ii. The success of _____ project depends on effective teamwork. (Article)	**2**	**3**	**2**
	c).	Construct one example sentence for any two of the following idiomatic expressions. i. A piece of cake ii. Break the ice iii. Once in a blue moon iv. Spill the beans	**3**	**3**	**2**
	d).	Write about the use of skimming technique in reading the texts.	**4**	**3**	**2**
	e).	Dos and don'ts of social media messaging.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Develop an essay on the strategies that can help a person sound smarter, more confident, and articulate when communicating.	**1**	**3**	**5**
	b).	Write a dialogue between a student and a librarian about borrowing a reference book and renewing the library membership.	**1**	**3**	**5**
		OR			
3.		Construct a paragraph on "The Importance of Time Management" in about 100–120 words and list out the cohesive devices used in the paragraph.	**1**	**3**	**10**
		UNIT-2			
4.	**a).**	Explain how eco-conscious negotiation strategies can be applied to address an environmental issue and promote a sustainable future.	**2**	**3**	**5**
	b).	Employ appropriate verb forms to fill in the blanks. i. By next year, she ___ (finish) her degree. ii. The train ___ (arrive) at the station just now.	**2**	**3**	**5**

		iii. I ___ (study) English for three years. iv. When I called him, he ___ (sleep). v. Listen! Someone ___ (play) the guitar.			
		OR			
5.		Apply appropriate grammatical rules and correct any **TEN** of the following sentences. i. Each of the engineers are responsible for testing the software. ii. The equipment in the laboratory were not functioning properly. iii. He discussed about the advantages of artificial intelligence. iv. The results of the experiment shows a significant improvement. v. The engineer suggested to use a different algorithm for the project. vi. If the system will fail, the backup server will be activated. vii. He is resembling his father. viii. The film show began before we reached the theatre. ix. Neither the project manager nor the team members was aware of the technical issue. x. She is an university student pursuing engineering. xi. Rahul went to school despite of having a fever. xii. I prefer chocolate milkshake than vanilla ice cream.	**2**	**3**	**10**
		UNIT-3			
6.	**a).**	Illustrate the appropriate ethical guidelines to address a challenge arising from the use of artificial intelligence or other disruptive technologies.	**3**	**3**	**5**
	b).	Construct sentences by substituting the given phrases with suitable one-word for any **FIVE** of the following. i. One who loves mankind. ii. One who cannot be defeated iii. Hobby of collecting coins. iv. Something that cannot be avoided v. A speech made without preparation vi. A person who spends money wastefully vii. A person who designs and develops software viii. A person who loves and collects	**3**	**3**	**5**
		OR			
7.	**a).**	Construct a formal letter to the Editor of a newspaper expressing your concern about the increasing use of plastic in your city. Apply appropriate letter-writing conventions and suggest possible solutions.	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Develop an essay on how innovative business practices can be applied to address challenges faced by modern startups.	**4**	**3**	**5**

	b).	Analyze the following hints to construct a formal email by applying appropriate writing techniques, tone, and structure. **Student ---> HR Manager:** New training/internship program -latest technology- useful – my career-want to be included- participate keenly-learn and implement- benefit of the company.	**4**	**3**	**5**
		OR			
9.	**a).**	Analyze the given passage and write answers to the following questions. Technology has transformed the way students learn and access information. Digital platforms, online courses, and educational applications allow learners to study at their own pace and explore subjects beyond the classroom. Engineering students, in particular, can use simulations, virtual laboratories, and collaborative tools to understand complex concepts more effectively. However, technology should complement traditional learning rather than completely replace it. Excessive dependence on digital resources may reduce face-to-face interaction and critical thinking. Therefore, students should use technology responsibly and balance digital learning with classroom discussions, practical activities, and independent study. Questions i. How has technology transformed the learning process? ii. What are the advantages of digital platforms and online courses? iii. How can engineering students benefit from simulations and virtual laboratories? iv. Why should technology not completely replace traditional learning? v. What are the possible disadvantages of excessive dependence on digital resources? vi. What does the word "complement" mean in the context of the passage? vii. Find a word from the passage that means "working together." viii. Find a word from the passage that means "too much or more than necessary." ix. What three activities does the passage recommend balancing with digital learning? x. What is the central idea of the passage?	**4**	**3**	**10**
		UNIT-5			
10.	**a).**	Demonstrate how core values can be integrated into ethical decision-making and socially responsible leadership.	**5**	**3**	**5**
	b).	Construct the following sentences as directed in the brackets.	**5**	**3**	**5**

		i. She has written many novels. (Add Question Tag) ii. The report was prepared by the committee. (Convert it into Active Voice) iii. He worked hard to pass the examination. (Convert it into a Complex Sentence) iv. Very few engineering colleges in India are as reputed as IIT Madras. (Convert it into the Superlative Degree) v. Smrithi is the cleverest girl in the class. (Change it into Comparative Degree)			
		OR			
11.	**a).**	Develop the content to be used in a blog on 'Social Media'.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2610002
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		LINEAR ALGEBRA & CALCULUS			
		(Common to All Branches of Engineering)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	KL	M
1.	(a)	Find the value of *k* such that rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2	1	3	2
	(b)	Find the singular values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	2	3	2
	(c)	Find the value of *c* in [0, 4] for $f(x) = x(x-2)(x-3)$ by using Lagrange's mean value theorem.	3	3	2
	(d)	Evaluate $\underset{\substack{x\to\infty \\ y\to 2}}{Lt}\left(\frac{xy+1}{x^2+2y^2}\right)$	4	3	2
	(e)	Evaluate $\int_0^2 \int_0^4 (x^2+y^2)\,dxdy$	5	3	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	(a)	Determine the rank of the matrix $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$ by reducing it to echelon form.	1	3	5
	(b)	Find the inverse of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Gauss – Jordan method.	1	3	5
		OR			
3.	(a)	For what values of a and b do the equations $x+2y+3z=6$, $x+3y+5z=9$ and $2x+5y+az=b$ have (i) No solution (ii) An unique solution (ii) Infinitely many solutions	1	3	5
	(b)	Solve the equations $20x+y-2z=17$, $3x+20y-z=-18$and $2x-3y+20z=25$ by Gauss-Seidel iteration method.	1	3	5
		UNIT-2			
4		Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	2	3	10

		OR			
5.		Reduce $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to sum of squares by linear transformation and hence find the matrix of the transformation, Nature, Signature and index of the quadratic form.	2	3	10
		UNIT-3			
6.	(a)	Verify Rolle's theorem for $\frac{\sin x}{e^x}$ in $(0, \pi)$.	3	3	5
	(b)	Expand $f(x) = \cos x$ by Maclaurin's theorem with Lagrange's form of remainder.	3	3	5
		OR			
7.	(a)	Find the appropriate value of c used in Cauchy's Mean-Value theorem for $f(x) = \ln x$ in [1, e]	3	3	5
	(b)	Expand $\tan^{-1}x$ in powers of $(x-1)$ upto fourth degree term.	3	3	5
		UNIT-4			
8.	(a)	If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	4	3	5
	(b)	Expand $f(x,y) = e^y \log(1+x)$ in powers of x and y using Maclaurin's series.	4	3	5
		OR			
9.	(a)	If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	4	3	5
	(b)	Discuss the maxima and minima of $f(x,y) = x^3y^2(1-x-y)$	4	3	5
		UNIT-5			
10.	(a)	Evaluate $\int\int xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by changing it in to polar coordinates	5	3	5
	(b)	By the change of order of integration evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	5	3	5
		OR			
11.		Evaluate $\iiint_V \frac{dxdydz}{x^2+y^2+z^2}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610101
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		PHYSICS FOR CIVIL ENGNIEERS			
		(For CE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	a)	State and explain the principle of Superposition.	**1**	**2**	**2**
	b)	Explain the procedure for obtaining Miller indices of a (hkl) plane.	**2**	**2**	**2**
	c)	Explain any one method for detecting ultrasonic waves.	**3**	**2**	**2**
	d)	Distinguish between pulse-echo and transmission methods.	**4**	**3**	**2**
	e)	Describe the working principle of AFM.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.		Illustrate the Newton's rings experiment and explain how it can be used to determine the wavelength of monochromatic light.	**1**	**3**	**10**
		OR			
3.		Derive the conditions for principle maximum, minima, and secondary maxima in Fraunhofer diffraction due to a single slit and draw the intensity distribution curve.	**1**	**3**	**10**
		UNIT-2			
4.		Illustrate SC,BCC and FCC crystal structures and compute their packing fractions.	**2**	**3**	**10**
		OR			
5.		Apply the working principle of SEM to examine the surface of a fractured material.	**2**	**4**	**10**
		UNIT-3			
6.		Describe the factors affecting reverberation time and explain their importance in auditorium design.	**3**	**3**	**10**
		OR			
7.	**a)**	Describe the generation of ultrasonic waves using the magnetostriction method.	**3**	**3**	**5**
	b)	Explain the applications of ultrasonics in various fields.	**3**	**3**	**5**
		UNIT-4			
8.		Derive an expression for the acceptance angle and numerical aperture of an optical fiber.	**4**	**3**	**10**
		OR			

9.	**a)**	Explain the pulse-echo method of ultrasonic testing and describe its application.	**4**	**3**	**5**
	b)	Describe the principle of ultrasonic testing and explain its working for detecting internal defects.	**4**	**4**	**5**
		UNIT-5			
10.	**a)**	Explain the important applications of Nanomaterials.	**5**	**3**	**5**
	b)	Describe chemical vapour deposition method for synthesis of nanomaterials.	**5**	**3**	**5**
		OR			
11.		Analyze how STM and AFM are used for studying the surface characteristics of nanomaterials.	**5**	**4**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		SUSTAINABLE DEVELOPMENT: PRINCIPLES, PRACTICES AND SDGS			
		(For CE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	State the three dimensions of sustainable development.	1	1	2
	b).	List the 5 Ps of Agenda 2030.	2	1	2
	c).	Define carbon footprint and mention one major source of carbon emissions from buildings.	3	1	2
	d).	Define Mission LiFE? State one responsible lifestyle practice promoted by it.	4	1	2
	e).	List any four Indian policies or initiatives that support implementation of the SDGs.	5	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	Explain the concept, principles and significance of sustainable development.	1	2	5
	b).	Discuss the environmental impacts of construction and of cement production, and the role of civil engineers in promoting sustainability.	1	2	5
		OR			
3.	a).	Explain the evolution from the Millennium Development Goals (MDGs) to the Sustainable Development Goals (SDGs) under Agenda 2030.	1	2	5
	b).	Explain the role of civil engineers in sustainable development.	1	2	5
		UNIT-2			
4.	a).	Explain the significance of the 5 Ps: People, Planet, Prosperity, Peace and Partnership.	2	2	5
	b).	Explain about the targets and indicators in measuring SDG progress.	2	2	5
		OR			
5.	a).	Explain the interlinkages between different Sustainable Development Goals with suitable examples.	2	2	5
	b).	Describe how progress on SDGs is measured, with reference to the global indicator framework and the NITI Aayog SDG India Index.	2	2	5
		UNIT-3			

6.	a).	Explain about the biodiversity and natural resource conservation with examples.	3	2	5
	b).	Explain water security and sustainable water management with reference to rainwater harvesting, water conservation and reuse.	3	2	5
		OR			
7.	a).	Explain the waste management and circular economy in construction.	3	2	5
	b).	Mention one local environmental problem and suggest a sustainable solution.	3	3	5
		UNIT-4			
8.	a).	Discuss the major causes of poverty and hunger and suggest suitable measures to improve food security.	4	2	5
	b).	Explain the role of education and lifelong learning in achieving sustainable development.	4	2	5
		OR			
9.	a).	Explain responsible consumption and responsible lifestyles with reference to Mission LiFE.	4	2	5
	b).	Explain the role of technology and innovation in sustainable development with suitable examples	4	2	5
		UNIT-5			
10.	a).	Explain the role of government in implementing the SDGs.	5	2	5
	b).	List and explain any four Indian initiatives supporting sustainable development.	5	2	5
		OR			
11.	a).	Explain SDG localization? Explain its importance.	5	2	5
	b).	Prepare a simple SDG-based action plan for a sustainability problem in a college campus.	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Find the integrating factor of $\frac{dy}{dx} - \frac{3y}{x} = x$	**1**	**3**	**2**
	b).	Solve$(D^2 - 4D + 4)y = 0$	**2**	**3**	**2**
	c).	Obtain a P.D.E from $z = ax + by + c$	**3**	**2**	**2**
	d).	Determine $\nabla\emptyset$, where $\emptyset = \sqrt{x^2 + y^2 + z^2}$	**4**	**3**	**2**
	e).	Find div $\bar{F}$, where $\bar{F} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$	**5**	**3**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Solve $y \log y \, dx + (x - \log y)dy = 0$	**1**	**3**	**5**
	b).	Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$	**1**	**3**	**5**
		OR			
3.	**a).**	If the air is maintained at 30°C and the temperature of the body cools from 100°C to 70° C in 15 minutes, determine the time at which the temperature of the body will be 40° C.	**1**	**3**	**5**
	b).	Find the orthogonal trajectories of the family of curves $r^2 = a^2 Sin2\theta$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Solve $(D-2)^2 y = 8(e^{2x} + x^2)$	**2**	**3**	**5**
	b).	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$	**2**	**3**	**5**
		OR			
5.	**a).**	Solve simultaneous equations $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that $x = 2$ and $y=0$ when $t = 0$	**2**	**3**	**5**
	b).	The damped LCR circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$ where L, C, R are positive constants. Find the critical resistance.	**2**	**3**	**5**
		UNIT-3			

6.	**a).**	Form the partial differential equation by eliminating arbitrary function from $f(x^2+y^2+z^2, z^2-2xy)=0$.	**3**	**3**	**5**
	b).	Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$	**3**	**3**	**5**
		OR			
7.		Solve $\left(D^2-3DD'+2D'^2\right)z=e^{x+2y}+Sin(x+2y)$	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Find the directional derivative of $f = xy+yz+zx$ in the direction of vector $\bar{i}+2\bar{j}+2\bar{k}$ at the point (1,2,0).	**4**	**3**	**5**
	b).	If $\overrightarrow{F} = 3x^2\overrightarrow{\iota}+5xy^2\overrightarrow{J}+5xyz^3\vec{k}$,then find $\nabla(\nabla.\vec{F})$	**4**	**3**	**5**
		OR			
9.	**a).**	Prove that $(y^2-z^2+3yz-2x\ )\vec{\imath}+(\ 3xz+2xy\ )\overrightarrow{J}+$ (3xy-2xz+2z)$\vec{k}$ is solenoidal.	**4**	**3**	**5**
	b).	Find the angle between the surfaces $xy^2z=3x+z^2$ and $3x^2-y^2+2z=1$ at the point $(1,-2,1)$	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Find the work done by the force field$\vec{F}=(2x+y^2)\vec{\imath}+xz\overrightarrow{J}+xyz\vec{k}$in moving a particle along a straight line from (0,0,0) to (2,1,3).	**5**	**3**	**5**
	b).	Evaluate $\oint_c\{(x^2y\,dx+x^2dy\}$ where C is a circle $x^2+y^2=a^2$ by using Green's theorem.	**5**	**3**	**5**
		OR			
11.		Verify Gauss – divergence theorem for$\vec{F}=2xz\overrightarrow{\iota}+yz\vec{J}+z^2\vec{k}$ where S is an upper half surface of the sphere$x^2+y^2+z^2=a^2$	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620101
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		CHEMISTRY FOR CIVIL ENGINEERS			
		(For CE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Classify the ions responsible for total hardness of water.	1	3	2
	b).	Relate the water–cement ratio to its importance in concrete.	2	3	2
	c).	Apply the Pilling–Bedworth rule to metal oxidation.	3	3	2
	d).	Illustrate the concept of composite materials with a suitable example.	4	3	2
	e).	Demonstrate the concept of green chemistry with a suitable example.	5	3	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Demonstrate the EDTA titration procedure for determining the hardness of water.	1	3	5
	b).	Apply the process of reverse osmosis used for desalination of water.	1	3	5
		OR			
3.	**a).**	Employ the concepts of pH, cation exchange capacity, and electrical conductivity to assess the chemical properties of soil.	1	3	5
	b).	Illustrate the interaction between clay particles and water, including adsorbed, capillary and gravitational water.	1	3	5
		UNIT-2			
4.	**a).**	Determine the major Bogue compounds of Portland cement and their roles in cement hydration.	2	3	5
	b).	Demonstrate the hydration of cement and the formation of C–S–H, calcium hydroxide and ettringite.	2	3	5
		OR			
5.	**a).**	Interpret the concepts of alkali–aggregate reaction and sulphate attack to identify the causes of concrete deterioration.	2	3	5
	b).	Apply the concept of water–cement ratio to assess the strength and work ability of concrete.	2	3	5
		UNIT-3			
6.	**a).**	Classify chemical corrosion and electrochemical corrosion and sketch the mechanism of electrochemical corrosion.	3	3	5

	b).	Interpret carbonation-induced and chloride-induced corrosion of reinforcing steel in concrete.	3	3	5
		OR			
7.	**a).**	Use the concepts of galvanic corrosion, pitting corrosion and concentration-cell corrosion to explain steel deterioration.	3	3	5
	b).	Utilize cathodic protection by sacrificial anode and impressed-current methods.	3	3	5
		UNIT-4			
8.	**a).**	Classify addition and condensation polymerization with suitable examples.	4	3	5
	b).	Differentiate thermoplastics and thermosetting polymers with suitable examples and construction applications.	4	3	5
		OR			
9.	**a).**	Utilize the concept of composite materials to classify them based on their matrix materials and identify their components.	4	3	5
	b).	Relate glass-fibre, carbon-fibre and aramid-fibre reinforced polymers to their applications in structural strengthening.	4	3	5
		UNIT-5			
10.	**a).**	Employ the principles of green chemistry in sustainable construction with suitable examples	5	3	5
	b).	Relate the utilization of fly ash, GGBS and steel slag in construction to their benefits.	5	3	5
		OR			
11.	**a).**	Illustrate the formation of geopolymer concrete, including precursors, alkali activators and the polymerization reaction.	5	3	5
	b).	Discuss the use of plastic waste in roads, including the process, advantages and limitations.	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620007
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		Computational Thinking and Problem Solving using python			
		(Common to CE & ME)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**K1**	**M**
1.	**a).**	List the five pillars of computational thinking.	**1**	**1**	**2**
	b).	Explain the Conditional Statement in Python.	**2**	**2**	**2**
	c).	Differentiate between mutable and immutable data structures.	**3**	**3**	**2**
	d).	Explain code reusability.	**4**	**2**	**2**
	e).	Define exception handling.	**5**	**1**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Describe the problem-solving process using decomposition, pattern recognition, abstraction, and algorithmic thinking.	**1**	**2**	**5**
	b).	Explain the standard flowchart symbols and their purposes with suitable examples.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain Python variables, data types, operators, and input/output with examples.	**1**	**2**	**5**
	b).	Construct and explain a flowchart to determine whether a given number is even or odd.	**1**	**4**	**5**
		UNIT-2			
4.	**a).**	Develop a Python program to generate the multiplication table of a given number using a loop.	**2**	**4**	**5**
	b).	Explain how loops can be viewed as an application of pattern recognition in computational thinking.	**2**	**2**	**5**
		OR			
5.	**a).**	Develop a Python program to find the sum of all even numbers within a given range.	**2**	**4**	**5**
	b).	Explain how conditional statements and loops can be combined to solve a problem involving multiple cases and repetitions.	**2**	**2**	**5**
		UNIT-3			

6.	**a).**	Analyze a Python program involving lists, tuples, and dictionaries and trace the changes in the data structures.	**3**	**4**	**5**
	b).	Examine different data structures for representing key-based and ordered information in a given application.	**3**	**4**	**5**
		OR			
7.	**a).**	Explain a Python program that uses strings, lists, tuples, and dictionaries to organize and process a given dataset.	**3**	**2**	**5**
	b).	Develop a Python program to perform suitable operations on a given string.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain recursion and the role of a base case in a recursive algorithm.	**4**	**2**	**5**
	b).	Explain Python modules and built-in libraries with suitable examples.	**4**	**2**	**5**
		OR			
9.	**a).**	Compare iterative and recursive approaches for solving a problem in terms of algorithmic efficiency.	**4**	**3**	**5**
	b).	Discuss responsible AI-assisted coding practices when developing Python programs.	**4**	**2**	**5**
		UNIT-5			
10.	**a).**	Develop a Python program that handles an appropriate runtime error using exception handling.	**5**	**3**	**5**
	b).	Explain the basic features and applications of NumPy, Pandas, and Matplotlib.	**5**	**2**	**5**
		OR			
11.	**a).**	Examine the role of visualization in identifying patterns in real-world datasets.	**5**	**3**	**5**
	b).	Describe how the concepts learned across all five units can be integrated into an end-to-end capstone project.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		ENGINEERING GEOLOGY			
		(For CE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	What is weathering? Mention types of weathering.	1	2	2
	b).	What is mineral? Give two examples.	2	2	2
	c).	What is strike and dip in structural geology?	3	2	2
	d).	Define cone of depression?	4	2	2
	e).	List types of dams.	5	2	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	Explain the term "geology" and discuss branches of geology and their significance in civil engineering.	1	2	5
	b).	Discuss the scope and importance of geology in civil engineering.	1	2	5
		OR			
3.	a).	Illustrate types of weathering by various geological agents.	1	2	5
	b).	Explain geological work of running water.	1	2	5
		UNIT-2			
4.	a).	Describe physical properties of minerals.	2	2	5
	b).	Explain feldspar group of minerals.	2	2	5
		OR			
5.	a).	Illustrate types of rocks with examples.	2	2	5
	b).	Discuss engineering properties of rocks.	2	2	5
		UNIT-3			
6.	a).	Define fold and explain parts of a fold.	3	2	5
	b).	Illustrate types of folds with neat sketches.	3	2	5
		OR			
7.	a).	Summarise types of faults with neat sketches and explain their significance in civil engineering.	3	2	5
	b).	Explain role of unconformities in geological investigations.	3	2	5
		UNIT-4			

8.	a).	Discuss methods of ground water exploration.	4	2	5
	b).	Explain landslides and discuss their preventive methods.	4	2	5
		OR			
9.	a).	Discuss the precautions of building constructions in seismic areas	4	2	5
	b).	Elucidate electrical resistivity method for identification of bedrock.	4	2	5
		UNIT-5			
10.	a).	Explain types of dams and their significance.	5	2	5
	b).	Illustrate geological investigations for dam site suitability.	5	2	5
		OR			
11.	a).	Summarize the types of tunnels.	5	2	5
	b).	Explain the influence of Geology for successful Tunneling.	5	2	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620103
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		Foundations of Quantum Technologies			
		(For CE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain photoelectric effect.	**1**	**2**	**2**
	b).	State and explain Heisenberg's uncertainty principle.	**2**	**2**	**2**
	c).	Define coherent states.	**3**	**1**	**2**
	d).	State and explain the first law of thermodynamics.	**4**	**2**	**2**
	e).	Define the complexity class PSPACE?	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.		Derive Hamilton's equations from the Hamiltonian function H(q,p,t).	**1**	**3**	**10**
		OR			
3.		Apply Hamilton's equations to a one-dimensional simple harmonic oscillator and obtain its equations of motion.	**1**	**3**	**10**
		UNIT-2			
4.		State the time-dependent Schrödinger equation and explain how it describes the time evolution of a quantum state.	**2**	**2**	**10**
		OR			
5.		Explain about Dirac Bra Ket notation, Hermitian Operators, and Unitary Transformations in Quantum mechanics.	**2**	**2**	**10**
		UNIT-3			
6.	**a)**	State and explain the No-Cloning Theorem.	**3**	**2**	**5**
	b)	Describe about Quantum Superposition and Entanglement.	**3**	**2**	**5**
		OR			
7.		Using the Schrödinger equation, obtain the allowed energy eigenvalues of a particle in a one-dimensional box.	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Compare classical and quantum harmonic oscillators using Gibbs statistics.	**4**	**3**	**5**
	b).	Discuss the significance of zero-point energy in a quantum harmonic oscillator.	**4**	**2**	**5**
		OR			

9.		Explain the statistical behavior of a quantum harmonic oscillator using the Gibbs principle.	**4**	**2**	**10**
		UNIT-5			
10.	**a).**	Define Shannon entropy mathematically. Explain how it quantifies the uncertainty in a discrete random variable?	**5**	**2**	**5**
	b).	What is quantum decoherence? Explain how interaction with the environment can reduce the coherence of a quantum state and affect quantum information.	**5**	**2**	**5**
		OR			
11.		Explain the purpose of post-quantum cryptography and why it is needed in the context of quantum computing.	**5**	**2**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

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